

3. Dada $A \in \mathbb{R}^{m \times n}$, defina $B(\lambda) = (A^T A + \lambda I)^{-1} A^T$ con $\lambda > 0$. Demuestre que si $p = \min\{m, n\}$, $\sigma_1 \geq \dots \geq \sigma_r > \sigma_{r+1} = \dots = \sigma_p = 0$ son los valores singulares de A y $A^\dagger$ es su pseudo inversa, entonces

$$\|B(\lambda) - A^\dagger\|_2 = \frac{\lambda}{\sigma_r(\sigma_r^2 + \lambda)}.$$

Concluya que $B(\lambda) \rightarrow A^\dagger$ si $\lambda \rightarrow 0^+$.

Sea $A = U \hat{\Sigma} V^T$, U, V ortogonales

y $\hat{\Sigma} = \begin{bmatrix} \Sigma & 0 \\ 0 & 0 \end{bmatrix}$ donde, $\Sigma = \begin{bmatrix} \sigma_1 & & & \\ & \sigma_2 & & \\ & & \ddots & \\ & & & \sigma_r \end{bmatrix}$

Vamos por parte:

• $A^T A + \lambda I = V \hat{\Sigma}^T \underbrace{U^T U}_{=I} \hat{\Sigma} V^T + \lambda I =$

$$= V \hat{\Sigma}^T \hat{\Sigma} V^T + \lambda I \quad (1)$$

• $(A^T A + \lambda I)^{-1} A^T = (V \hat{\Sigma}^T \hat{\Sigma} V^T + \lambda I)^{-1} V \hat{\Sigma}^T U^T$
 $= V (\hat{\Sigma}^T \hat{\Sigma} + \lambda I)^{-1} \underbrace{V^T V}_{=I} \hat{\Sigma}^T U^T$

$$= V (\hat{\Sigma}^T \hat{\Sigma} + \lambda I)^{-1} \hat{\Sigma}^T U^T \quad (2)$$

• $A^\dagger = V \begin{bmatrix} \Sigma^{-1} & 0 \\ 0 & 0 \end{bmatrix} U^T \quad (3)$

Ahora:

$$\|B(\lambda) - A^{\dagger}\|_2 = \|(A^T A + \lambda I)^{-1} A^T - A^{\dagger}\|_2$$

Reemplazamos
 (1), (2) y (3) $\rightarrow$

$$= \left\| V (\hat{\Sigma}^T \hat{\Sigma} + \lambda I)^{-1} \hat{\Sigma}^T U^T - V \begin{bmatrix} \Sigma^{-1} & 0 \\ 0 & 0 \end{bmatrix} U^T \right\|_2$$

$$= \left\| V \left((\hat{\Sigma}^T \hat{\Sigma} + \lambda I)^{-1} \hat{\Sigma}^T - \begin{bmatrix} \Sigma^{-1} & 0 \\ 0 & 0 \end{bmatrix} \right) U^T \right\|_2$$

$$= \left\| (\hat{\Sigma}^T \hat{\Sigma} + \lambda I)^{-1} \hat{\Sigma}^T - \begin{bmatrix} \Sigma^{-1} & 0 \\ 0 & 0 \end{bmatrix} \right\|_2 =$$

$$= \left\| \begin{bmatrix} \frac{\sigma_1}{\sigma_1 + \lambda} & & & \\ & \frac{\sigma_2}{\sigma_2 + \lambda} & & \\ & & \ddots & \\ & & & \frac{\sigma_r}{\sigma_r + \lambda} & & 0 \dots 0 \end{bmatrix} - \begin{bmatrix} 1/\sigma_1 & & & \\ & 1/\sigma_2 & & \\ & & \ddots & \\ & & & 1/\sigma_r & & 0 \dots 0 \end{bmatrix} \right\|_2$$

$$= \left\| \begin{bmatrix} \frac{\lambda}{\sigma_1(\sigma_1^2 + \lambda)} & & & \\ & \frac{\lambda}{\sigma_2(\sigma_2^2 + \lambda)} & & \\ & & \ddots & \\ & & & \frac{\lambda}{\sigma_r(\sigma_r^2 + \lambda)} & & 0 \dots 0 \end{bmatrix} \right\|_2 =$$

$$= \frac{\lambda}{\sigma_r(\sigma_r^2 + \lambda)}$$

